JAKE RYAN

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MASTER CV — internal reference only. Not for direct submission; trim to a role-specific version (see `software-developer/` or `robotics-engineer/`) before sending.

PROFESSIONAL SUMMARY

Computer Science graduate with cross-disciplinary experience spanning full-stack web development and robotics/embedded systems. Comfortable across the stack, from REST APIs and React frontends to ROS-based control loops and sensor fusion on physical hardware. Track record of shipping full-stack products, running human-subject research studies, and publishing/presenting technical work.

EXPERIENCE

Undergraduate Research Assistant June 2020 – Present
Texas A&M University *College Station, TX*

- Built a REST API with FastAPI and PostgreSQL to ingest and serve data from learning management systems
- Shipped a full-stack app (Flask, React, PostgreSQL, Docker) to analyze GitHub activity across student teams
- Designed interactive dashboards to visualize classroom collaboration patterns
- Built data pipelines to log and analyze sensor/interaction data from research systems
- Containerized services with Docker to keep experiments reproducible across lab machines

Information Technology Support Specialist Sep. 2018 – Present
Southwestern University *Georgetown, TX*

- Communicated with managers to set up campus computers used across departments
- Assessed and troubleshot computer problems brought by students, faculty and staff
- Maintained upkeep of computers, classroom equipment, and 200+ printers across campus
- Automated device provisioning workflows, cutting manual setup time for new lab computers
- Maintained and repaired lab hardware, including embedded and classroom automation equipment

Artificial Intelligence Research Assistant May 2019 – July 2019
Southwestern University *Georgetown, TX*

- Explored methods to generate video game dungeons based off of *The Legend of Zelda*
- Developed a game in Java to test the generated dungeons
- Contributed 50K+ lines of code to an established codebase via Git
- Conducted a human subject study to determine which video game dungeon generation technique is enjoyable
- Wrote an 8-page paper and gave multiple presentations on-campus
- Presented virtually to the World Conference on Computational Intelligence

PROJECTS

Gitlytics | *Python, Flask, React, PostgreSQL, Docker* June 2020 – Present

- Built a full-stack app with Flask serving a REST API consumed by a React frontend
- Implemented GitHub OAuth to pull repository data per authenticated user
- Used Celery + Redis to run data-sync jobs asynchronously without blocking the UI

Autonomous Line-Following Rover | *ROS, C++, Python, Raspberry Pi* 2021

- Built a ROS-based rover with a PID control loop for line-following using a camera and IR sensor array
- Implemented a basic obstacle-avoidance node using ultrasonic sensor fusion
- Tuned control parameters in simulation (Gazebo) before deploying to hardware

Simple Paintball | *Spigot API, Java, Maven, TravisCI, Git* May 2018 – May 2020

- Developed a Minecraft server plugin to entertain kids during free time for a previous job
- Published plugin to websites gaining 2K+ downloads and an average 4.5/5-star review
- Implemented continuous delivery using TravisCI to build the plugin upon new release
- Collaborated with Minecraft server administrators to suggest features and get feedback about the plugin

Technical Skills

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript/TypeScript, HTML/CSS, R
Web Frameworks: React, Node.js, Flask, FastAPI, JUnit, WordPress, Material-UI
Robotics & Control: ROS/ROS2, Gazebo, PID control, path planning, sensor fusion (basic SLAM concepts)
Hardware: Raspberry Pi, Arduino, IR/ultrasonic sensors, basic embedded I/O
Developer Tools: Git, Docker, TravisCI, Google Cloud Platform, Linux, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse
Data & Analysis: pandas, NumPy, Matplotlib, MATLAB (basic), statistical study design

Education

Southwestern University — Georgetown, TX
Bachelor of Arts in Computer Science, Minor in Business — *Aug. 2018 – May 2021*

Blinn College — Bryan, TX
Associate's in Liberal Arts — *Aug. 2014 – May 2018*

Publications & Talks

- **Dungeon Generation for Player Enjoyment**, 8-page paper; presented at the World Conference on Computational Intelligence (2019)